

TVPCI-R: JUST TRANSITION RETURN CHAIN

Bristol One City: Community Wealth Return Analysis

The TVPCI-R measures how much of the One City investment and benefit recirculates to Bristol's most deprived communities. TVPCI-R = 50.7 (Failure zone). B_chain = 47.0 (severe imbalance). The gap between the primary delivery chain (TVPCI 81.3%) and the community return chain (50.7%) is the defining structural failure of the Bristol system. All values PROVISIONAL.

81.3

TVPCI (primary chain balance)

50.7

TVPCI-R (community return score)

47.0

B_chain (holon balance)

+30.6

Delta_chain (value flowing out, not circulating back)

Framework: TVPCI, TVPCI-R, and B_chain

In the Tholonic framework, the primary delivery chain (TVPCI) measures D-C balance: how well the institutional architecture (D) is matched by actual delivery (C). The return chain (TVPCI-R) measures a different question: of the value generated by the delivery chain, how much circulates back to the communities that most need it? For Bristol, this is the "just transition" dimension: the degree to which the decarbonisation transition also reduces inequality and improves outcomes for the 53,100 residents in the most deprived 10% nationally.

D ROLE: DEFINITION**TVPCI (Primary Chain)**

How well does the institutional delivery chain balance its requirements (D) against its outputs (C)? Measures the coherence of the city's delivery system as a whole.

81.3 (Good)**C ROLE: CONTRIBUTION****TVPCI-R (Return Chain)**

How much of the investment and benefit generated by the delivery chain recirculates to deprived communities? Measures the just transition dimension: the "ecological return" of social equity.

50.7 (Failure)**N ROLE: NEGOTIATION****B_chain (Holon Balance)**

The emergent coherence of the whole system: how well the primary chain and the return chain are in balance. A high B_chain means the city is both delivering and returning value proportionally. $B_chain = 100 \times \exp(-2 \times |TVPCI - TVPCI-R| / \max(TVPCI, TVPCI-R))$.

47.0 (Severe imbalance)

Delta_chain = TVPCI - TVPCI-R = +30.6 (positive: value is flowing forward through the institutional chain but not circulating back to deprived communities)

TVPCI (primary chain)

81.3%

TVPCI-R (return chain)

50.7%

Per-Phase Return Scores (R_p) and Omega Weights

All R_p scores and omega weights are PROVISIONAL. R_p measures how observable and accountable each phase's community return is. Omega weights are proportional to community impact potential at each phase and must sum to 1.0. Zone thresholds: Coherent R_p ≥ 80; Stressed 61.8-79; Failure 38.2-61.8; Breakdown < 38.2.

P0: VISION

R_p = 62

STRESSED

ω = 0.08

Plan names equity explicitly. QoL disaggregated by deprivation. But no enforcement of equity requirements in mandate text.

P1: GOVERNANCE

R_p = 55

FAILURE

ω = 0.08

VOSCUR (voluntary sector) and TUC represented on Governance Board. Community voice is present but outweighed by institutional anchors. Co-design of missions limited.

P2: STRATEGY

R_p = 60

FAILURE

ω = 0.10

City Leap has £61.5m social value target incl. local supply chain. BCAI accessible from £5. But strategies are technically-authored, not co-designed with deprived communities.

P3: CAPITAL

R_p = 48

FAILURE

ω = 0.18

~£30m of £61.5m social value delivered. Lawrence Weston community energy. BCAI at £2m. Most capital flows to institutional partners, not deprived communities directly.

P4: INFRASTRUCTURE

R_p = 65

STRESSED

ω = 0.22

Solar at schools, housing retrofit focused on social housing, LED lighting in public buildings. Direct benefit to deprived communities

P5: COMMUNITY UPTAKE ⚠

R_p = 35

BREAKDOWN

ω = 0.25

750 BCAI investors from 470k population. Hartcliffe, Filwood, Lawrence Hill not visibly benefiting. Scope 3 unaddressed.

P6: OUTCOMES

R_p = 40

FAILURE

ω = 0.09

QoL survey disaggregated by deprivation. But life expectancy gap 10.7yr not improving. Mental wellbeing gap persistent. Outcomes

is real but geographically narrow.

The primary community interface is essentially absent.

do not yet reflect community wealth return.

TVPCI-R Calculation

PHASE	R_P	Ω (OMEGA)	Ω × R_P	ZONE
P0: Vision and Mandate	62	0.08	4.96	STRESSED
P1: Governance	55	0.08	4.40	FAILURE
P2: Strategy Design	60	0.10	6.00	FAILURE
P3: Capital Deployment	48	0.18	8.64	FAILURE
P4: Infrastructure Delivery	65	0.22	14.30	STRESSED
P5: Community Uptake Δ	35	0.25	8.75	BREAKDOWN
P6: Outcome Measurement	40	0.09	3.60	FAILURE
TOTAL		1.00	50.65	FAILURE

TVPCI-R = Σ(ω × R_p) = 50.7 | B_chain = 100 × exp(-2 × |81.3 - 50.7| / 81.3) = 100 × exp(-0.754) = 47.0

Community Impact Assessment: Who Receives the Return?

The omega weights above reflect community impact potential: Phase 5 carries the highest weight (0.25) because it is the direct community interface, and Phase 4 (0.22) because infrastructure physically enters deprived communities. The collapse of R_p at Phase 5 (35, Breakdown) combined with its high omega weight is the primary driver of the low TVPCI-R.

COMMUNITY DIMENSION	CURRENT STATUS	RETURN CHAIN SIGNAL	PHASE(S)
Financial participation	750 BCAI investors from 470,000 population (0.16%)	Breakdown: financial return is not reaching deprived communities	P3, P5
Health inequality	10.7yr male life expectancy gap (most vs least deprived, 2021-23)	Failure: health returns not reflected in outcome data despite significant investment	P5, P6

COMMUNITY DIMENSION	CURRENT STATUS	RETURN CHAIN SIGNAL	PHASE(S)
Mental wellbeing	24.4% poor mental wellbeing (most deprived) vs 11.5% (least deprived)	Failure: wellbeing gap persistent, not narrowing	P5, P6
Physical infrastructure access	Solar at schools, housing retrofit in social housing, LED lighting in public buildings	Stressed: tangible but geographically narrow; 30 deprived wards not fully covered	P4
Energy cost benefit	Pay-as-you-save model designed but not yet deployed at scale	Failure: the financial benefit of the energy transition has not yet reached households in fuel poverty	P3, P5
Employment (local supply chain)	£30m of £61.5m social value target delivered; £50m local supply chain target	Failure: approximately 49% of 5yr social value target delivered at midpoint; local SME pipeline still developing	P3, P4
Scope 3 / consumption equity	No structured city-wide scope 3 reduction programme at household level	Breakdown: the consumption emissions of deprived communities (often lower per capita) are not being tracked or supported	P5, P6

Priority Just Transition Interventions (Bond-Linked)

The following interventions are the highest-priority actions to close the TVPCI-TVPCI-R gap. Each is structured to be measurable, phase-specific, and bondable (i.e. capable of serving as a KPI or SPT in a Sustainability-Linked Bond or Local Climate Bond tranche).

PHASE	INTERVENTION	DESCRIPTION	EXPECTED R_P IMPACT	BOND/FINANCE LINK
P5 (primary)	Net-Zero Neighbourhood demonstrators in 3 deprived wards	Co-design and fund net-zero retrofit of c.1,000 properties each in Hartcliffe/Withywood, Filwood, and Lawrence Hill (Bristol's three most deprived wards). Blended finance: SLB (senior) + City Funds (mezzanine) + pay-as-	R_p P5: 35 → approximately 55-60 (+20-25 pts). Omega = 0.25. TVPCI-R impact: +5-6 pts overall.	SLB SPT: number of properties retrofitted in IMD top-10% wards. Observation date: 2027 (1,000 props), 2030 (3,000 props).

PHASE	INTERVENTION	DESCRIPTION	EXPECTED R_P IMPACT	BOND/FINANCE LINK
		you-save (junior). Living Places Earth model.		
P3 (capital)	Deprived-ward BCAI tranche with council co-contribution	Third BCAI tranche at £1 minimum entry, ISA-eligible, with a council match-contribution for Universal Credit recipients. Target: 10,000 investors from the 53,100 in the most deprived decile. Raise BCAI total to £5-7m.	R_p P3: 48 → approximately 58-62 (+10-14 pts). Omega = 0.18. TVPCI-R impact: +2-3 pts.	SLB SPT: number of BCAI investors from IMD top-10% postcode areas. Verification: Abundance Investment investor postcode data.
P5 and P6	Scope 3 ward-level consumption tracking	Partner with Cabot Institute to add a consumption-based emissions supplement to the annual QoL Survey. Create ward-level scope 3 tracking for 34 wards, disaggregated by deprivation decile. Establishes the data layer needed for scope 3 SPTs in a future SLB.	R_p P5: +8-10 pts (new measurable output). R_p P6: +10-12 pts (data quality improvement). Combined TVPCI-R impact: +3-4 pts.	Pre-condition for scope 3 SLB KPI. Bond-linkable once methodology established and first data published.
P3 and P5	SLB issuance with life expectancy gap SPT	Issue a £50-100m Sustainability-Linked Bond with the male life expectancy gap (most vs least deprived wards) as a Social SPT alongside the environmental KPIs. 25 bps step-up if gap exceeds 10.0 years at 2030 observation. TrueValue N-D-C scoring provides annual independent verification of all SPTs.	R_p P3: +15 pts (institutional capital with equity accountability). TVPCI-R systemic impact: +4-6 pts when all KPIs are linked to community return. B_chain: 47.0 → approximately 58-62.	Direct SLB SPT: male life expectancy gap at 2027 and 2030 observation dates. Coupon step-up of 25 bps if missed. Verification: PHOF (A02a), ONS data.

Calculation Reference

R_p = community return score for phase p (0-100): how observable and accountable the phase's community return is. Ω_p = phase weight (proportional to community impact potential; sums to 1.0). $TVPCI_R = \sum(\Omega_p \times R_p)$. $B_{chain} = 100 \times \exp(-2 \times |TVPCI - TVPCI_R| / \max(TVPCI, TVPCI_R))$. Zone thresholds (phi-derived): Coherent $R_p \geq 80$ | Stressed 61.8-79 | Failure 38.2-61.8 | Breakdown < 38.2 .

METRIC	VALUE	ZONE	INTERPRETATION
TVPCI (primary chain)	81.3	GOOD	The institutional delivery chain is functioning well in aggregate, with Phase 5 as the exception.
TVPCI-R (return chain)	50.7	FAILURE	Community wealth return is in the Failure zone. No phase achieves Coherent status.
B_chain (holon balance)	47.0	BREAKDOWN	Severe imbalance between primary delivery and community return. The system is not in sustainable equilibrium.
Delta_chain (TVPCI - TVPCI-R)	+30.6	Imbalanced	Positive delta: value is flowing forward through the institutional chain but not recirculating back. This is the structural definition of an unjust transition.
Weakest return phase	P5: $R_p = 35$ (Breakdown)	BREAKDOWN	Community and household uptake is the primary bond intervention candidate for the return chain.